

Magnetoelastic wave in Ferromagnetic Thin-film Mediated by Dipolar Interaction

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Abstract. Magnetoelastic coupling mediated by magnetic dipolar ~~interactions~~ interaction is theoretically investigated in ~~ferromagnetic thin films a ferromagnetic thin film~~ under an in-plane magnetic field. We ~~construct a microscopic framework~~ develop a theoretical description that incorporates dipolar fields derived from Maxwell's equations in the presence of elastic ~~deformation~~ deformations. The resulting coupled equations of motion predict hybridization between magnetostatic and Lamb waves. Numerical calculations for a yttrium iron garnet (YIG) film reveal ~~hybridization of modes and~~ anti-crossings ~~of 0.1 to several MHz~~ in the dispersion relations, with hybridization gaps ranging from 0.1 to several MHz.

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Magnetoelastic coupling has attracted considerable attention because of its potential for manipulating magnetic properties via mechanical degrees of freedom [1, 2]. At the microscopic level, magnons and phonons are coupled ~~through-via~~ spin-orbit interaction, ~~which gives rise to magnon-phonon coupling~~ [1, 3, 4]. In contrast, at longer wavelengths ~~the coupling is mainly mediated by the magnetic dipolar interaction. In this regime, magnetostatic waves interact with elastic waves. Experimental studies of such dipolar-interaction-mediated magnetoelastic coupling have recently progressed owing to the development of~~ magnetic dipolar interactions also play an important role for the coupling. Recent experimental advances, particularly those based on interdigital transducers (IDTs) [5, 6], ~~which enable precise measurements of acoustic excitations, have enabled coherent excitation and precise characterization of acoustic waves in magnetic thin films. However, a~~ [5, 6], allowing direct access to magnetoelastic coupling in this regime. While continuum theories describing magnetostatic [7, 8, 9] and elastic waves [10, 11, 12, 13] in this regime have been developed separately, a unified theoretical framework that ~~consistently incorporates the dipolar interaction in magnetoelastic dynamics has not yet been established, because of its long-ranged feature and its singularity when two dipoles comes~~ captures their coupling mediated by dipolar interactions remains lacking, primarily due to the long-range nature of the dipolar interaction and its singular behavior at short distances.

In this Letter, ~~by circumventing the above difficulties using the Green function method, we construct a general formalism that incorporates the dipolar interaction and describe the resulting coupling between magnetostatic and elastic waves in ferromagnetic thin films. we extend the Green-function method developed by Kalinikos and Slavin [14] to incorporate dipolar interactions in the presence of elastic deformations. This approach allows us to quantitatively evaluate the magnetoelastic coupling mediated by dipolar interactions.~~

Our system is schematically shown in Fig. 1. We consider a ferromagnetic thin film that is infinitely extended in the xy plane and has a finite thickness L in the z direction, with the origin taken at the middle of the film. ~~The film~~ It is assumed to be isotropic both magnetically and elastically. An in-plane external magnetic field $\mathbf{H}_0$ is applied, resulting

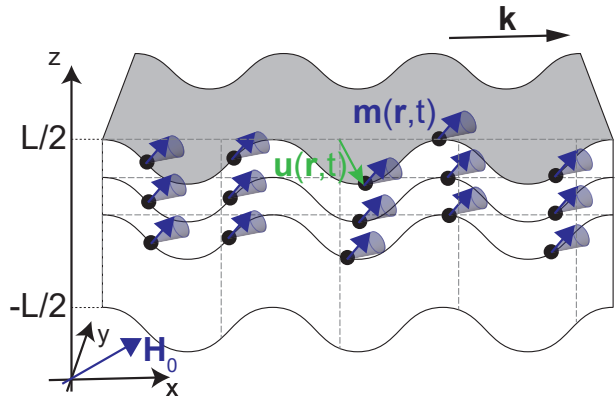


Figure 1. Schematic illustration of the system under consideration. A ferromagnetic thin film with thickness L in the z direction and infinite extent in the x and y directions is considered. Magnetic moments precess around an axis parallel to the in-plane external magnetic field $\mathbf{H}_0$. The elastic wave changes the positions of the magnetic moments and their separations, resulting in a modification of the dipolar interaction among them.

in an equilibrium saturation magnetization $\mathbf{M}_0$ parallel to $\mathbf{H}_0$.

We decompose the magnetic field $\mathbf{H}$ and the magnetization $\mathbf{M}$ as $\mathbf{H} = \mathbf{H}_0 + \mathbf{h}(\mathbf{r}, t)$ and $\mathbf{M} = \mathbf{M}_0 + \mathbf{m}(\mathbf{r}, t)$, respectively, where $\mathbf{r} = (x, y, z)$ is the Euler coordinate, and study propagating magnetostatic waves of small deviations from the equilibrium magnetization $\mathbf{m}(\mathbf{r}, t)$. The system also supports elastic waves described by the displacement $\mathbf{u}(\mathbf{r}, t)$. The deformation modulates the distance between magnetic dipoles, so that the ~~dipole-dipolar~~ field $\mathbf{h}(\mathbf{r}, t)$ generated by $\mathbf{m}$ is influenced by $\mathbf{u}$, leading to a magnetoelastic wave. We take the x axis parallel to the propagation direction, and assume that waves are plane waves and the system is uniform in the y direction, ~~where we write $\mathbf{r} = (x, y, z)$.~~ Then, $\mathbf{m}$ and $\mathbf{u}$ can be written as

$$\mathbf{m}(\mathbf{r}, t) = \mathbf{m}(z)e^{i(kx - \omega t)}, \quad \mathbf{u}(\mathbf{r}, t) = \mathbf{u}(z)e^{i(kx - \omega t)}, \quad (1)$$

where $\mathbf{m}(z), \mathbf{u}(z) \in \mathbb{C}^3$. Hereafter, all fields are treated as complex quantities, and physical observables are given by their real parts. We study the coupled dynamics of these waves up to first order in these quantities. Here, we focus exclusively on the case of an in-plane magnetic field. The case of an arbitrary external magnetic field will be presented elsewhere.

We start from the basic equations describing the magnetic dipolar interaction, namely Maxwell's equations. When elastic deformation is present, the magnetic dipoles deviate from their equilibrium positions $\mathbf{r}$ to $\mathbf{R} := \mathbf{r} + \mathbf{u}(\mathbf{r}, t)$. The Maxwell's equations are then rewritten in terms of this ~~new~~ Lagrange coordinate as

$$\nabla_{\mathbf{R}} \cdot (\mathbf{H} + \mathbf{M}) = 0, \quad \nabla_{\mathbf{R}} \times \mathbf{H} \approx 0, \quad (2)$$

where in the second equation the magnetostatic approximation is applied and the time derivative of the electric field is discarded. This approximation is justified when the ~~length scale is sufficiently long so that the dipolar interaction is dominant~~ steady current is absent and the characteristic frequency ω and length scale L of the magnetization dynamics satisfy the quasi-static condition $\omega L/c \ll 1$, so that retardation effects and electromagnetic radiation can be ignored [15]. By the plane-wave assumption (1), the dipolar field can also be written as $\mathbf{h} = \mathbf{h}(z)e^{i(kx - \omega t)}$. We rewrite Eq. (2) in terms of these quantities in the ~~original~~ Euler coordinate $\mathbf{r}$. We note that we need to pay attention to the conservation of the magnetization in a unit volume [16] yielding $\mathbf{M}(\mathbf{R}) = \mathbf{M}(\mathbf{r})/\det(J(\mathbf{r}))$, where $J(\mathbf{r})$ is a Jacobian matrix (see Supplementary data [17] for details). We get

$$\nabla_{\mathbf{r}} \cdot (\mathbf{h}(\mathbf{r}) + \mathbf{m}(\mathbf{r})) = \mathbf{M}_0 \cdot \nabla_{\mathbf{r}} (\nabla_{\mathbf{r}} \cdot \mathbf{u}(\mathbf{r})), \quad (3)$$

$$\nabla_{\mathbf{r}} \times \mathbf{h}(\mathbf{r}) = 0, \quad (4)$$

where $\nabla_{\mathbf{r}}$ stands for derivatives with respect to the $\mathbf{r}$ coordinate. Because the surface of the film is also distorted as in Fig. 1, the boundary conditions used to solve these equations are also modified by the presence of $\mathbf{u}$ as

$$h_x^{\text{in}} - h_x^{\text{out}} = \underline{\underline{M_{0,z} \frac{\partial u_z}{\partial x}}}, \quad (5)$$

$$h_z^{\text{in}} + m_z - h_z^{\text{out}} = M_{0,x} \frac{\partial u_z}{\partial x}, \quad (6)$$

at $z = \pm L/2$, where ~~$h^{\text{in/out}}$ stands-stand~~ stands-stand for the dipolar fields inside and outside the film, respectively. Hereafter, lower indices indicate ~~component in the $\mathbf{r}$ -components in the Euler~~ components in the Euler coordinate. Following the Green tensor method as in Ref. [14], the dipolar field satisfying Eqs. (3) and (4) under the boundary conditions (5) and (6) is given as (see Supplementary data [17] for details)

$$\mathbf{h}(z) = \int_{-\frac{L}{2}}^{\frac{L}{2}} dz' (G^m(z, z') \mathbf{m}(z') + G^u(z, z') \mathbf{u}(z')), \quad (7)$$

where

$$G^m(z, z') = \begin{pmatrix} -G_P(z, z') & 0 & -iG_Q(z, z') \\ 0 & 0 & 0 \\ -iG_Q(z, z') & 0 & G_{P'}(z, z') \end{pmatrix}, \quad (8)$$

$$G^u(z, z') = -ikM_{0,x}G^m(z, z'),$$

$$G_P(z, z') = \frac{k}{2}e^{-k|z-z'|}, \quad (9)$$

$$G_Q(z, z') = \text{sgn}(z - z')G_P(z, z'), \quad (10)$$

$$G_{P'}(z, z') = G_P(z, z') - \delta(z - z'). \quad (11)$$

This dipolar field couples the magnetostatic wave and the elastic wave in the film. We note that in the present setup there is no coupling when the equilibrium magnetization is parallel to the y axis, ~~i.e., when the external magnetic field is applied in that direction. Otherwise, the two waves are coupled via $\mathbf{h}$.~~

Next, we examine the effect of magnetoelastic coupling on the dynamics of the magnetostatic and elastic waves. In the absence of this coupling, the dynamics of the magnetization $\mathbf{m}$ up to first order is described by the linearized Landau-Lifshitz equation

$$\dot{\mathbf{m}} = -\gamma\mu_0 (\mathbf{m} \times \mathbf{H}_0 + \mathbf{M}_0 \times \mathbf{h}), \quad (12)$$

where γ is the gyromagnetic ratio and μ_0 is the vacuum magnetic permeability. This equation can be derived from a Lagrangian

$$L^m[\mathbf{m}, \dot{\mathbf{m}}, \mathbf{m}^*, \dot{\mathbf{m}}^*] = \int dz \Re \left[\frac{1}{4M_0^2\gamma} \mathbf{M}_0 \cdot (\mathbf{m} \times \dot{\mathbf{m}}^*) - \frac{\mu_0}{4} \left(\mathbf{h}^* \cdot \mathbf{h} + \frac{H_0}{M_0} \mathbf{m}^* \cdot \mathbf{m} \right) \right], \quad (13)$$

where the asterisk shows complex conjugate. We can verify this by constructing the Euler-Lagrange equation with respect to $\mathbf{m}^*$ from this Lagrangian. On the other hand, the dynamics of the displacement field $\mathbf{u}$ is described by the Navier-Cauchy equation

$$\rho \ddot{\mathbf{u}} = \mu \nabla^2 \mathbf{u} + (\mu + \lambda) \nabla (\nabla \cdot \mathbf{u}), \quad (14)$$

where ρ is the mass density, μ and λ are Lamé constants. The Lagrangian for this equation is

$$L^u[\mathbf{u}, \dot{\mathbf{u}}, \mathbf{u}^*, \dot{\mathbf{u}}^*] = \int dz \Re \left[\frac{\rho}{4} \dot{\mathbf{u}}^* \cdot \dot{\mathbf{u}} + \frac{1}{4} \partial_i \sigma_{ij} u_j^* \right], \quad (15)$$

where $\sigma_{ij} = \mu(\partial_i u_j + \partial_j u_i) + \lambda \delta_{ij} \nabla \cdot \mathbf{u}$ is the stress tensor. The total Lagrangian of the coupled system is given by the sum of the two Lagrangians, Eqs. (13) and (15). In this case, $\mathbf{h}$ is no longer only a functional of the magnetization, but also a functional of the displacement field $\mathbf{u}$. As a result, a new volume force term ~~$-\mu_0 \mathbf{f}$~~ $\mathbf{f}$ appears in the Navier-Cauchy equation (14), where the force density $\mathbf{f}$ is given by (see Supplementary data [17] for details of the derivation)

$$f_i(z) := \underline{\underline{\mu_0 \frac{\delta h_j^*}{\delta u_i^*} h_j}} = \underline{\underline{-ikM - ik\mu_0 M_{0,x} h_i(z)}}. \quad (16)$$

Physically, this force can be understood as a Maxwell stress acting on the magnetic dipoles, caused by the change in the dipolar field induced by the displacement

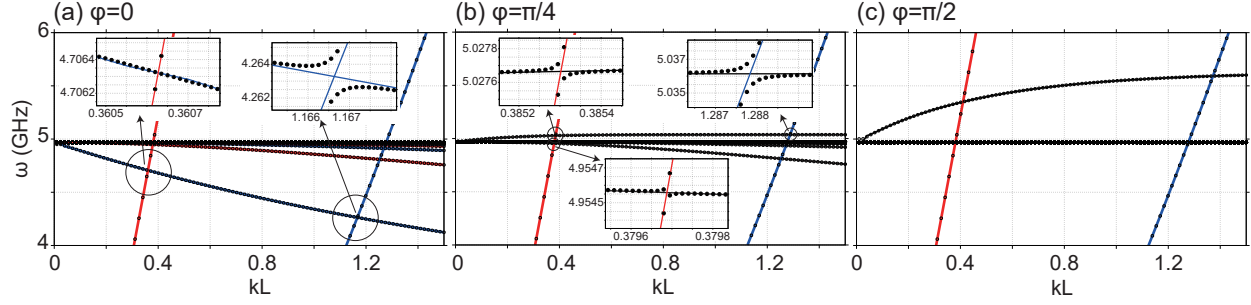


Figure 2. Dispersion curves of the coupled waves. Red and blue curves represent ~~symmetric and antisymmetric modes of the uncoupled waves, including magnetostatic backward volume waves (BVWs) with C_{2x} eigenvalues +1 and Lamb waves -1, while black respectively.~~ Black curves represent magnetostatic modes that are ~~neither symmetric nor antisymmetric not eigenstates of the C_{2x} operator.~~ Black dots Markers show the dispersion of the coupled waves obtained from numerical calculations. The external magnetic field is applied (a) parallel to the propagation direction, $\mathbf{H}_0 \parallel \mathbf{k}$ ($\varphi = 0$), (b) at $\varphi = \pi/4$, and (c) perpendicular to the propagation direction, $\mathbf{H}_0 \perp \mathbf{k}$ ($\varphi = \pi/2$). Insets show magnified views of representative band crossings. In (a), $\varphi = 0$, anticrossings appear between BVW and Lamb modes only when they have the same C_{2x} ~~eigenvalues eigenvalue~~ (same color). When the C_{2x} symmetry of the system is broken (b), both the magnetostatic surface mode and the BVW couple to Lamb modes. ~~The bottom-center inset of panel (b) shows the hybridization between the lowest-frequency BVW mode and the first Lamb wave mode.~~

field. The coupled dynamics of the magnetoelastic wave is then given by

$$\begin{cases} \dot{\mathbf{m}} = -\gamma\mu_0(\mathbf{m} \times \mathbf{H}_0 + \mathbf{M}_0 \times \mathbf{h}) \\ \rho\ddot{\mathbf{u}} = \mu\nabla^2\mathbf{u} + (\mu + \lambda)\nabla(\nabla \cdot \mathbf{u}) - \mathbf{f}. \end{cases} \quad (17)$$

In order to examine the dynamical properties of the coupled waves described by the above equations of motion, we numerically calculate the dispersion relation in the following. Under the assumption (1), the time ~~derivative-derivatives~~ on the left-hand side of the equations ~~becomes become~~ $-i\omega$. The second ~~equation~~ of Eqs. (17) contains spatial derivatives of $\mathbf{u}$ and requires boundary conditions to ~~solve itbe solved~~. Since we consider a film in vacuum, we impose the ~~tension-free stress-free~~ condition at the surfaces $z = \pm L/2$ as

$$\sigma_{xz}|_{z=\pm L/2} = \sigma_{yz}|_{z=\pm L/2} = \sigma_{zz}|_{z=\pm L/2} = 0. \quad (18)$$

Without the coupling, it is known that a type of elastic wave called a Lamb wave appears under these conditions. Since the system in equilibrium is symmetric with respect to the $z = 0$ plane, the Lamb wave modes are either ~~symmetric or antisymmetric with respect to the sign change of the variable even or odd in z .~~ Numerically, we need to solve the equations under the boundary conditions (18). For example, the finite element method naturally incorporates these conditions.

On the other hand, the magnetostatic waves without the coupling to elastic waves in this geometry were studied in detail by Damon and Eshbach [2][8]. They analytically showed that when the external magnetic field is parallel to the propagation direction of the wave ($\mathbf{H}_0 \parallel \mathbf{k}$), the system has infinitely many bulk modes called backward volume waves (BVWs); ~~which are either symmetric or antisymmetric.~~ When

the magnetic field is in-plane but perpendicular to the propagation direction ($\mathbf{H}_0 \perp \mathbf{k}$), a surface mode appears that is localized at one of the surfaces.

For the calculation of the dispersion, it is convenient to introduce another coordinate ~~(r_1, r_2, r_3)~~ system, where the original ~~coordinate one~~ (x, y, z) is rotated around the z axis so that ~~the r_1 one of the axis~~ is parallel to the external magnetic field. Then, the r_1 -component of $\mathbf{m}$ is zero ~~along this direction~~ since $\mathbf{M}_0 \cdot \mathbf{m} = 0$ from the constraint $|\mathbf{M}| = \text{const.}$ $\dashv$

Now we consider the coupled equations (17). Using the finite element method, we discretize the equations in real space while satisfying the boundary conditions, which leads to a generalized eigenvalue problem for the eigenvalue ω . By solving this problem, we obtain the dispersion curves shown in Fig. 2. The parameters are taken to be those of Yttrium-Iron-Garnet (YIG): $\rho = 5.17 \text{ g/cm}^3$, $\mu = 7.81 \times 10^{-8} \text{ kg} \cdot \text{GHz}^2/\text{m}$, $\lambda = 10.94 \times 10^{-8} \text{ kg} \cdot \text{GHz}^2/\text{m}$ [18], $\gamma = 29.8 \text{ GHz/T}$, and $\mu_0 M_0 = 0.178 \text{ T}$ [19]. The vacuum permeability is $\mu_0 = 4\pi \times 10^{-11} \text{ T}^2\text{m}/(\text{kg} \cdot \text{GHz}^2)$. We set the film thickness to ~~$L = 500 \mu\text{m}$~~ $L = 500 \text{ nm}$ and the external magnetic field to $\mu_0 H_0 = 0.1 \text{ T}$.

We define the angle between the x axis and the external magnetic field as φ and write $\mathbf{H}_0 = H_0(\cos \varphi, \sin \varphi, 0)$. In Fig. 2(a), we plot the case $\varphi = 0$. In this case, ~~magnetostatic-BVWs appear, plotted as blue and red curves representing the antisymmetric and symmetric modes with respect to the twofold rotation about the x -axis (the system has a C_{2x}), with eigenvalues -1~~ rotation symmetry and thus, ~~the modes of magnetostatic BVWs and Lamb waves are eigenstates of the C_{2x} operation. States with C_{2x} eigenvalue -1 ($C_{2x}\mathbf{v}(z) = -\mathbf{v}(z)$, ($\mathbf{v} = \mathbf{m}, \mathbf{u}$) are plotted as blue curves, while those with +1~~ $\dashv$

respectively. Similarly, antisymmetric and symmetric Lamb waves with positive group velocities (positive slopes) are shown by blue and red curves as red curves. Curves with negative slopes are magnetostatic BVWs and positive ones are Lamb waves, respectively. The dots markers represent the dispersion of the coupled waves. The insets show magnified views of the band crossings between BVW-BVWs and Lamb waves. We observe that the magnetostatic and elastic waves couple and produce anti-crossings cause anti-crossing and produce hybridization gaps of several MHz only when the two modes have the same C_{2x} eigenvalues. This is because when $\mathbf{H}_0 \parallel \mathbf{k}$, the system possesses the two-fold rotation symmetry with respect to the x -axis C_{2x} and coupled modes needs to be either symmetric or anti-symmetric. Figure 2(b) shows the case $\varphi = \pi/4$. In this case, two types of magnetostatic modes coexist: BVWs and surface modes and the C_{2x} symmetry is broken waves. The insets show magnified views of the band crossings between the magnetostatic and Lamb modes. Small waves. Hybridization gaps at anti-crossings appear between all the modes as the two-fold rotation symmetry is broken system does not have any symmetries in this case. Finally, in Fig. 2(c), we plot the case $\varphi = \pi/2$. In this case here, a magnetostatic surface mode appears but does not couple to the Lamb modes waves. This is because $G^u = 0$, $\mathbf{f} = \mathbf{0}$ and Eq. (17) reduces to two independent equations. In other words, the magnetostatic modes are odd under the mirror reflection with respect to the xz plane, while the elastic modes are even, and therefore they do not couple.

From the magnitude of the hybridization gaps, the effective coupling strength g_{eff} is estimated to be on the order of a few MHz. On the other hand, assuming a typical Gilbert damping parameter $\alpha \approx 10^{-4}$ [20], a quality factor $Q \approx 10^3$, and operating frequencies ω in the GHz range, the corresponding magnon and phonon linewidths are estimated as $\gamma_m \sim \alpha\omega$ and $\kappa \sim \omega/Q$, respectively. Under these conditions, $g_{\text{eff}}^2/(\gamma_m\kappa) \sim 1$, suggesting that the system is close to the boundary of the strong-coupling regime [21]. Therefore, the hybridization gaps are expected to be experimentally resolvable.

In conclusion, we have constructed a theoretical framework to describe developed a theoretical description of magnetoelastic coupling mediated by magnetic dipolar interactions in ferromagnetic thin films. The dipolar field was derived from Maxwell's equations in the presence of elastic deformations, and the coupled equations of motion were obtained within the Lagrangian formalism. Numerical calculations were performed for a YIG thin film to obtain the dispersion relations of the coupled modes. We found mode hybridizations and resulting anti-crossings due to mode

hybridization, with gaps of several MHz in general. When the external in-plane magnetic field is parallel to the propagation of the waves, wave propagation, the system has C_2 symmetry, so only magnetostatic and elastic waves with the same parity- C_2 symmetry eigenvalues can hybridize. When they are perpendicular, couplings does do not occur. The formulation developed here provides a starting point for theoretical studies of dipolar-interaction-mediated magnetoelastic coupling in ferromagnetic thin films.

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